# Freight Rate Prediction

Validation approach, data-quality handling and the fixed December forecast

## 1. Summary

- **Task.** Predict posted_rate for 12,000 loads (Nov–Dec 2025) from 48,000 labelled loads (Jan–Oct 2025).
- **Model.** Ridge linear base on log(rate) plus a three-member LightGBM residual ensemble. Trained on 47,323 rows after excluding 677 corrupted labels.
- **Validation.** Forward chaining with a two-month horizon (train on everything before the window, predict the next two months), which is the same shape as the real task. Random k-fold was rejected because it leaks neighbouring days.
- **Result.** 1.87% MAPE and $44 MAE on the five forward folds, against 4.39% / $104 for a log-linear baseline and 8.33% / $168 for a median $/mile baseline (clean test rows).
- **Caveat.** The official metric is computed by Spotter and is not visible to me. All figures here are my own forward-validation numbers.

## 2. Data-quality findings and handling

| Issue | Evidence | Handling |
|---|---|---|
| Corrupted labels | About 670 rates are 2.4–5× too high or 0.2–0.4× too low for what the load implies (677 flagged by an out-of-fold model). Kept rows stay within ±0.20 log; flagged rows are all beyond ±0.80. | Excluded from training. Reported separately in evaluation (see 3.3). |
| Negative weights | 292 train and 145 validation rows. After taking the absolute value the range is 5,000–47,500, so these are sign flips. | abs(weight). |
| Missing values | weight: 300 train / 165 validation. market_index: 374 / 249. No day is entirely missing. | weight left as NaN plus a flag (LightGBM handles it). market_index filled with that day's median. |
| Unseen cities | 8 cities appear only in validation and touch 12.1% of validation rows. | City names are never used as features; only distance and coordinates. |
| Distance floor | Distance is floored at 70 miles (e.g. New Orleans–Shreveport is 7 miles apart by coordinates). | Kept; geometry features give the model both views. |
| Drifting market_index | Validation mean 0.93 vs training 1.08; strong weekly cycle (Thursday high, Sunday low). | Day-level median used as the market level plus the load's deviation from it. |

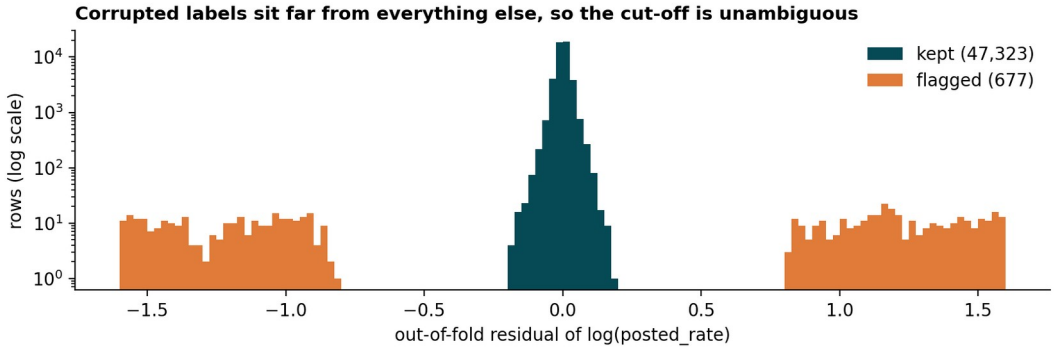



*Figure 1. Out-of-fold log residuals. The orange cluster is separated from the rest by a wide empty gap, so the threshold is not a judgement call.*

# 3. Train / test split and validation approach

## 3.1 Why not a random split

Rates depend on a slowly moving market level. The average residual of a model that ignores time has a lag-1 autocorrelation of 0.82 across days, so test rows from a random split have training rows from the same week sitting right next to them. Measured directly, random 5-fold gave 1.17% MAPE while forward chaining gave 1.87%. A random split would have understated the error by about 37%.

## 3.2 Forward-chaining design

The real task is to train on January–October and predict November–December. Each fold repeats that: train on all labelled days before a cut-off, predict the following two months. The primary folds start in May, June, July, August and September, so every fold has at least four months of history.

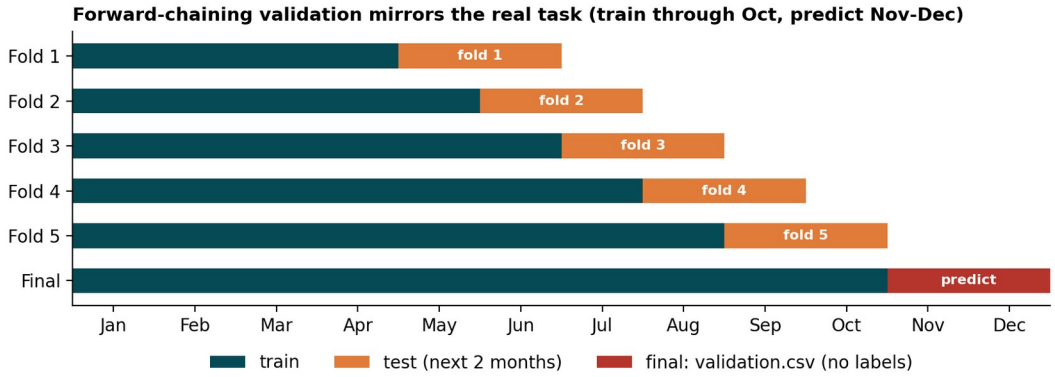



*Figure 2. Five forward folds plus the final fit. A test window never overlaps its training data.*

## 3.3 How the corrupted labels are treated in evaluation

Corrupted rows are excluded from training in every fold. Test metrics are reported two ways: on clean test rows (the model's real accuracy) and on all test rows including corrupted ones. Corrupted labels dominate the raw RMSE (roughly $630–$670 for every model), so differences between models only show up in the clean numbers and in raw MAE and MAPE.

## 3.4 Results

| Model (mean of 5 forward folds) | MAE $ | MAPE % | Raw MAE $ | Raw MAPE % |
|---|---|---|---|---|
| Median $/mile × distance (by equipment) | 168.2 | 8.33 | 220.7 | 10.46 |
| Log-linear (distance, equipment) | 104.3 | 4.39 | 157.9 | 6.60 |
| LightGBM: distance + equipment | 102.9 | 4.37 | 156.5 | 6.59 |
| + weight | 91.5 | 3.92 | 145.2 | 6.13 |
| + market index | 89.8 | 3.88 | 143.4 | 6.07 |
| + quote signal | 60.2 | 2.81 | 114.1 | 5.02 |
| + calendar (day of week, day of month) | 62.9 | 2.91 | 116.7 | 5.12 |
| + geometry (coordinates, haversine, circuity) | 56.2 | 2.48 | 110.1 | 4.70 |
| + time index | 51.3 | 2.16 | 105.2 | 4.43 |
| **Final: linear base + LightGBM ensemble** | **44.0** | **1.87** | **98.1** | **4.14** |

Final model by fold (MAPE %): May–Jun 1.55, Jun–Jul 1.51, Jul–Aug 1.98, Aug–Sep 2.34, Sep–Oct 1.99.

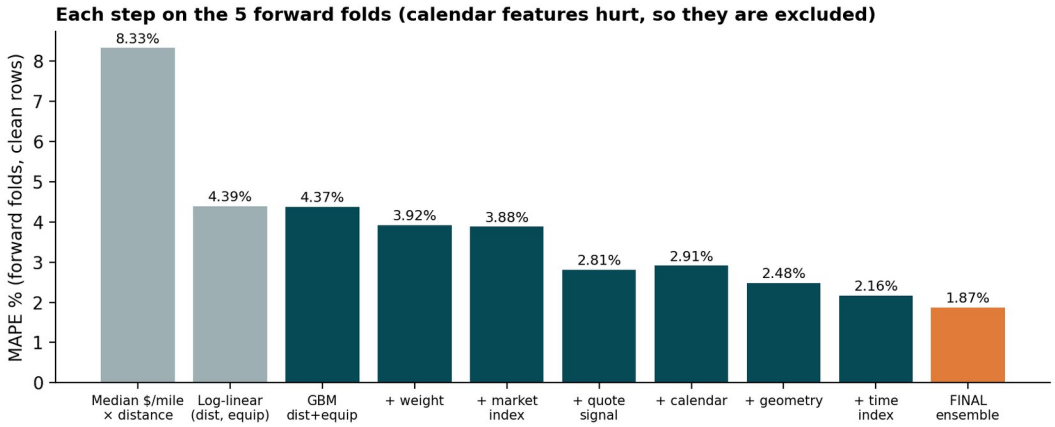



*Figure 3. Contribution of each feature group. Calendar features made things worse (market_index already carries the weekly cycle), so they are not in the final model.*

### 3.5 Robustness checks, including where it fails

- **Short-history stress folds.** With only two to three months of training (test windows starting March and April) error rose to 8.0% and 16.2% MAPE. In those windows quote_signal falls well outside the range seen in training and the model misreads it. The real run is not exposed to this: validation market_index (0.72–1.10) and quote_signal (1.23–2.90) both sit inside the training ranges (0.68–1.47 and 0.69–3.61), and the model trains on ten months.
- **Unseen cities.** I removed 8 random cities from training and tested on September–October. Rows touching those cities scored 1.98% MAPE with raw coordinates, versus 2.69% without raw coordinates and 2.80% with no geometry. Coordinates generalise to new cities, so they stay in.
- **quote_signal.** Its distribution shifts month to month and its within-month effect is small and inconsistent, so it partly works as a period fingerprint. I kept it because forward validation shows a real gain (3.88% to 2.81%), and the ensemble includes a member that sees only its day-level value. This is the main risk if November–December behaves differently from the months seen.

## 4. Why this model

- **Linear base on log(rate).** Distance, equipment, weight and market level act smoothly and multiplicatively, so a linear model extrapolates sensibly when the market level drifts.
- **LightGBM on the residual.** Learns interactions, regional effects and the slow time drift. A time index lets it carry the latest rate level forward, which cut forward error from 2.48% to 2.16%.
- **Three-member average.** Different feature views and regularisation (all features, day-level quote only, all features with stronger regularisation). In an earlier comparison the average scored 1.89% against 1.94–2.03% for the individual members.
- **Log target.** Rates span $57 to roughly $6,000 and errors are multiplicative. The prediction is the conditional median, which is the right target for MAE and robust to the label noise.

## 5. Fixed December prediction chart

The chart below was produced by the provided score.py from the completed december_chart_inputs.csv. Lane, equipment, distance and weight are fixed, so only the date varies.

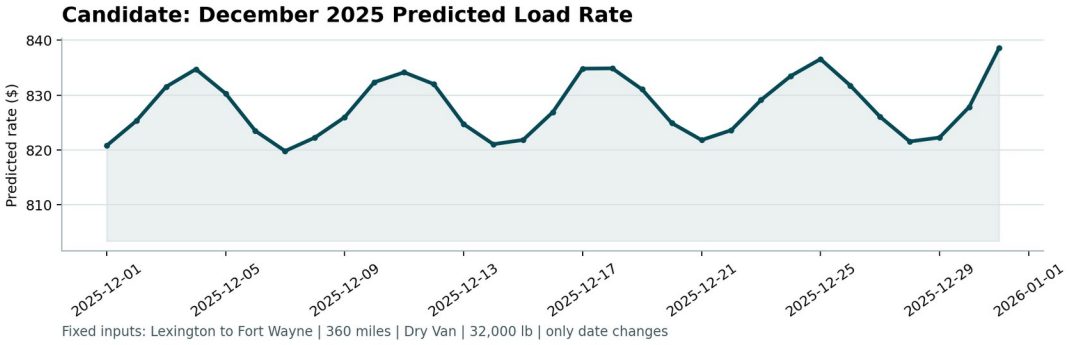



*Figure 4. candidate_december.png, unmodified scorer output. Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb.*

- **Inputs not in the file.** The December file has no market_index or quote_signal. I used the daily median of each across all loads on that date in validation.csv. These are features only, with no labels involved, and the load-level deviation is set to zero (a typical load that day).
- **What the chart shows.** Predicted rate sits between about $820 and $839 (about $2.30 per mile), with a weekly cycle of roughly 2% peak to trough that peaks around Thursday. The cycle comes from the weekly pattern in market_index, which is real in the data. The model has no information about December-specific demand beyond what market_index carries.
- **Level assumption.** The time index is flat beyond the last training day, so the model assumes the October rate level persists through December.

## 6. Limitations

- The scoring metric is hidden. If the hidden labels also contain about 1.4% corrupted rows, those rows will inflate any error metric regardless of model quality; I optimised for the conditional median.
- Forward validation covers May to October. I cannot test how the model behaves in a regime that never appears in the training year.
- Day-level market and quote medians for validation days are computed from validation features, which assumes those features are available at prediction time.